

## Abstract

Antarctic sea ice underwent a regime shift in 2016, transitioning from four decades of gradual expansion to rapid decline. Whether this shift activated a wave-ice positive feedback loop, in which retreating ice lengthens fetch, amplifies waves, and accelerates further ice loss, has remained untested with long-term reanalysis data. Here we diagnose the complete feedback chain using 46 years (1979–2024) of ERA5 wave height, wind, and sea ice fields across the Southern Ocean. We find that waves do erode the marginal ice zone: significant wave height Granger-causes marginal ice zone width ( $F = 8.35$ ,  $p < 0.001$ ). However, the hypothesized fetch feedback fails: open-water fraction does not predict wave height ( $p = 0.86$ ). Wave intensification is instead driven by strengthening westerlies ( $F = 3.59$ ,  $p = 0.007$ ), consistent with the positive trend in the Southern Annular Mode. The 2016 regime shift exposed a wider ocean surface to pre-existing, wind-driven wave energy, but did not amplify that energy through a self-reinforcing feedback. Among five major ice shelves, only the Ross Ice Shelf shows a significant reduction in sea ice buffer days. These results reframe the wave-ice interaction from a closed positive feedback to a one-directional, wind-mediated erosion mechanism operating on an already-shifted ice margin.

# Wind-driven wave erosion of Antarctic marginal ice zone intensifies after the 2016 sea ice regime shift

Author One<sup>1,\*</sup>, Author Two<sup>2</sup>

<sup>1</sup>Institution 1, City, Country

<sup>2</sup>Institution 2, City, Country

\*Corresponding author: email@institution.edu

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## 1 Introduction

Antarctic sea ice extent increased gradually from 1979 to 2014, then declined precipitously. The annual mean extent dropped by 11.5% between the pre-2016 and post-2016 periods, with the Pettitt change point test placing the regime shift at 2016 ( $p = 0.004$ )[1]. The decline was most pronounced in austral summer (December–February), when extent fell by 16.8%. This abrupt transition, unprecedented in the satellite era, has prompted the question of whether the Southern Ocean has crossed a tipping point into a new, self-reinforcing state of ice loss[2].

A leading hypothesis invokes a wave-ice positive feedback. Sea ice acts as a buffer that attenuates ocean swell before it reaches ice shelf margins and the interior ice pack[3, 4]. When ice retreats, the open-water fetch lengthens, wave energy increases, and stronger waves fragment and erode the marginal ice zone (MIZ), exposing yet more open water. This feedback loop, if active, would accelerate ice loss beyond what atmospheric forcing alone could produce and would have direct implications for ice shelf stability, since Massom et al.[3] demonstrated that ocean swell triggered the disintegration of the Larsen A, Larsen B, and Wilkins ice shelves following loss of their protective sea ice buffer.

The individual links of this chain have theoretical and observational support. Wave-ice interaction theory predicts that sea ice acts as a low-pass filter, preferentially attenuating short-period waves, with attenuation rates depending on ice concentration, thickness, and floe size[4]. Wave-triggered breakup generates lognormal floe size distributions in the MIZ[5], and fractured ice melts faster, creating a local feedback between fragmentation and concentration loss. Recent observations confirm that ocean waves drive Antarctic sea ice melting through flooding and mechanical pulverization[6], and that waves can propagate over 1000 km into the winter ice pack[7]. ICESat-2 altimetry has enabled direct observation of wave attenuation through the Antarctic MIZ[8].

What remains untested is whether these individual mechanisms close into a self-reinforcing loop at the basin scale, and specifically whether the 2016 regime shift activated or strengthened such a feedback. Addressing this requires simultaneous, long-term observations of sea ice concentration, wave height, wind forcing, and MIZ geometry across the full Southern Ocean, a combination that only reanalysis products can provide over multi-decadal timescales.

Here we use 46 years (1979–2024) of ERA5 monthly reanalysis to diagnose the complete wave-ice feedback chain in the Southern Ocean. We test four links: (1) whether sea ice loss increases open-water fraction (the fetch proxy), (2) whether increased open water amplifies significant wave height (SWH), (3) whether higher SWH erodes the MIZ, and (4) whether MIZ erosion feeds back to reduce total sea ice concentration. We also assess changes in ice shelf buffer conditions at five major Antarctic ice shelves.

## 2 Data and Methods

### 2.1 ERA5 reanalysis products

We use three ERA5 monthly-mean reanalysis products from the European Centre for Medium-Range Weather Forecasts (ECMWF)[9]:

- **Significant wave height (SWH)**, mean wave period (MWP), and mean wave direction (MWD) on the native  $0.5^\circ$  wave model grid, 1979–2024 (552 months). The ERA5 wave model (WAM) runs only in grid cells where sea ice concentration is below 30%, so SWH data are inherently limited to open and marginal waters.
- **10-m wind components** ( $u_{10}$ ,  $v_{10}$ ) on the  $0.25^\circ$  atmospheric grid, 1979–2024.
- **Sea ice concentration (SIC)** on the  $0.25^\circ$  atmospheric grid, 1979–2024. ERA5 SIC is derived from OSTIA satellite observations (from 1979 onward) assimilated into the ECMWF system.

All fields are restricted to the Southern Ocean domain  $40^\circ\text{S}$ – $75^\circ\text{S}$ , all longitudes. Wind speed is computed as  $|\mathbf{u}_{10}| = \sqrt{u_{10}^2 + v_{10}^2}$ .

### 2.2 NSIDC Sea Ice Index

We use the NSIDC Sea Ice Index v4.0[10] as an independent validation of Antarctic sea ice extent trends. Monthly Southern Hemisphere extent and area are available from 1979 to 2026. These satellite-derived estimates are independent of ERA5 SIC and provide a check on the timing and magnitude of the 2016 regime shift.

### 2.3 Regional definitions

We define four latitude bands for area-weighted spatial averaging:

- **Full Southern Ocean:**  $40^\circ\text{S}$ – $75^\circ\text{S}$
- **Near-MIZ zone:**  $55^\circ\text{S}$ – $65^\circ\text{S}$  (captures the mean position of the seasonal ice edge and MIZ)
- **ACC belt:**  $45^\circ\text{S}$ – $55^\circ\text{S}$  (largely ice-free, dominated by the Antarctic Circumpolar Current)
- **Deep ice edge:**  $65^\circ\text{S}$ – $75^\circ\text{S}$  (persistently ice-covered in winter)

The **marginal ice zone (MIZ)** is defined as the band where  $0.15 < \text{SIC} < 0.80$ . MIZ width is computed as the zonal-mean meridional extent of this band at each time step.

### 2.4 Open-water fraction as fetch proxy

Direct calculation of fetch (the unobstructed distance over which wind generates waves) requires resolving the geometry of ice floes and leads at each time step, which is not feasible with monthly-mean SIC fields. Instead, we use the **open-water fraction** (the fraction of grid cells with  $\text{SIC} < 0.15$  south of  $55^\circ\text{S}$ ) as a proxy for fetch. An increase in open-water fraction corresponds to a larger area of open water available for wave generation.

### 2.5 Change point detection

We apply the Pettitt test[11] to annual-mean anomaly time series of each variable to detect regime shifts. The Pettitt test is a non-parametric test for a single change point in the mean of a time series. The test statistic  $K$  and associated  $p$ -value are computed following the standard formulation. We report change point years, pre- and post-shift means, and significance levels.

## 2.6 Granger causality

We test directional predictability between pairs of monthly anomaly time series using Granger causality[12]. For each pair  $(x, y)$ , we test whether past values of  $x$  improve the prediction of  $y$  beyond what  $y$ 's own past values provide. We evaluate lag orders 1–6 months, selecting the optimal lag by AIC. We report the  $F$ -statistic and  $p$ -value of the restricted-vs-unrestricted model comparison.

The feedback chain is tested as a sequence of Granger causality links:

$$\text{SIC} \rightarrow \text{OW fraction} \rightarrow \text{SWH} \rightarrow \text{MIZ width} \rightarrow \text{SIC}$$

with wind speed ( $|\mathbf{u}_{10}|$ ) as an alternative driver of SWH.

## 2.7 Ice shelf buffer analysis

For five major Antarctic ice shelves (Ross, Filchner-Ronne, Amery, Larsen C, and Totten), we compute the number of months per year in which the mean SIC in a box encompassing the shelf front exceeds 0.15 (“buffer months”). A decrease in buffer months indicates increased exposure of the ice shelf front to open-ocean wave action.

## 3 Results

### 3.1 Confirmation of the 2016 sea ice regime shift

The NSIDC Sea Ice Index confirms a significant regime shift in 2016 (Pettitt  $p = 0.004$ ), with annual mean extent declining from 11.68 to 10.33 million  $\text{km}^2$  ( $-11.5\%$ ). The decline is strongest in austral summer (DJF:  $-16.8\%$ ,  $p = 0.002$ ) and weakest in winter (JJA:  $-5.7\%$ , not significant at  $p = 0.073$ ). The ERA5 SIC field confirms this timing: the area-weighted mean SIC in the ice zone ( $55^\circ$ – $75^\circ\text{S}$ ) shows a change point at 2016 ( $p = 0.021$ ), and the open-water fraction shifts synchronously ( $p = 0.013$ ). Before 2016, Antarctic sea ice extent increased at  $+0.022$  million  $\text{km}^2 \text{ yr}^{-1}$ ; after 2016, it declined at  $-0.311$  million  $\text{km}^2 \text{ yr}^{-1}$ .

### 3.2 Wave height trends are wind-driven, not fetch-driven

Significant wave height in the Southern Ocean has increased over the satellite era, but the change point occurs in 1993 ( $p = 0.001$ ), more than two decades before the sea ice regime shift. In the near-MIZ zone ( $55^\circ$ – $65^\circ\text{S}$ ), the SWH anomaly shows a marginal change point at 1993 ( $p = 0.055$ ), with mean anomaly shifting from  $-0.078$  m to  $+0.034$  m. The ACC belt ( $45^\circ$ – $55^\circ\text{S}$ ) shows the strongest SWH increase ( $p = 0.0001$ ), consistent with intensification of the westerly wind belt. Spatially, 17% of Southern Ocean grid points show a significant SWH increase between the pre-2016 and post-2016 periods ( $p < 0.05$ , two-sample  $t$ -test).

The correlation between near-MIZ SWH anomaly and sea ice extent anomaly is statistically significant but weak ( $r = 0.136$ ,  $p = 0.001$ ,  $n = 550$  months), indicating that sea ice conditions explain less than 2% of SWH variance.

### 3.3 The feedback chain is partially supported but not closed

Granger causality tests reveal which links in the hypothesized feedback chain hold and which fail (Table 1):

Three links are statistically supported: sea ice loss increases open-water fraction (the most robust link,  $F = 29.8$ ), wave height erodes MIZ width, and wind drives wave height. Two links fail: open-water fraction does not predict SWH ( $p = 0.86$ ), and MIZ width does not feed back to total SIC ( $p = 0.28$ ). The direct path from SIC to SWH also fails ( $p = 0.35$ ).

The MIZ width itself shows an earlier change point than sea ice concentration: 2007 ( $p < 0.001$ ) versus 2016 for SIC. This suggests that MIZ narrowing began before the overall ice decline and may reflect long-term changes in wave climate rather than a response to the 2016 shift.

Table 1: Granger causality tests for the wave-ice feedback chain. Optimal lag selected by AIC from 1–6 months. Significant results ( $p < 0.05$ ) in bold.

| Link                                            | $F$          | $p$              | Lag  | Status       |
|-------------------------------------------------|--------------|------------------|------|--------------|
| <b>SIC <math>\rightarrow</math> OW fraction</b> | <b>29.80</b> | <b>&lt;0.001</b> | 3 mo | Supported    |
| OW fraction $\rightarrow$ SWH                   | 0.32         | 0.864            | 4 mo | <b>Fails</b> |
| <b>SWH <math>\rightarrow</math> MIZ width</b>   | <b>8.35</b>  | <b>&lt;0.001</b> | 3 mo | Supported    |
| MIZ width $\rightarrow$ SIC                     | 1.30         | 0.275            | 3 mo | <b>Fails</b> |
| <b>Wind <math>\rightarrow</math> SWH</b>        | <b>3.59</b>  | <b>0.007</b>     | 4 mo | Supported    |
| SIC $\rightarrow$ SWH (direct)                  | 1.09         | 0.354            | 3 mo | Fails        |

### 3.4 Ice shelf buffer changes

Among the five ice shelves examined, only the Ross Ice Shelf shows a statistically significant reduction in sea ice buffer days (change point at 2009,  $p = 0.041$ , declining from 11.7 to 10.8 buffered months per year). Filchner-Ronne, Amery, Larsen C, and Totten show no significant changes in buffer conditions over the 1979–2024 period. The Ross result is consistent with documented retreat of fast ice in the western Ross Sea[3].

## 4 Discussion

### 4.1 Why the fetch feedback fails

The failure of the open-water fraction to predict SWH contradicts the intuitive expectation that more open water should generate larger waves. Three factors likely explain this result. First, the Southern Ocean has the longest fetches on Earth even with sea ice present, because the circumpolar geometry provides continuous open water between 40°S and the ice edge. A modest poleward retreat of the ice edge adds relatively little to the already-enormous fetch available for wave generation in the westerly wind belt. Second, ERA5 SWH at monthly timescales integrates swell propagating from remote storms across the full Southern Ocean basin, diluting any local fetch signal near the ice edge. Third, the ERA5 wave model does not compute wave generation in grid cells with SIC > 30%, creating a fundamental observational blind spot in the zone where fetch changes are most pronounced.

### 4.2 Wind-mediated erosion as the dominant mechanism

Our results support a simpler mechanism than the hypothesized feedback loop: the Southern Annular Mode (SAM) has trended positive over recent decades[13], strengthening the circumpolar westerlies and increasing wave energy across the Southern Ocean. The SWH change point at 1993, two decades before the sea ice shift, is consistent with this wind-driven intensification. The 2016 sea ice regime shift then exposed a wider band of ocean surface to this pre-existing, intensified wave field. Waves erode the MIZ (Granger  $p < 0.001$ ), but the wave energy itself is not amplified by the ice retreat. The interaction is one-directional: wind  $\rightarrow$  waves  $\rightarrow$  MIZ erosion, with sea ice loss as a permissive condition rather than a driver of wave intensification.

### 4.3 Implications for ice shelf stability

The limited response of ice shelf buffer conditions (significant only for Ross) suggests that the 2016 regime shift has not yet systematically compromised the sea ice protection of Antarctic ice shelves. However, the Ross Ice Shelf result, combined with the continued intensification of wind-driven wave energy, indicates that specific sectors may be approaching a threshold of buffer loss. If the SAM continues to trend positive and sea ice continues to decline, the coincidence of stronger

waves and reduced buffering could trigger the wave-driven calving mechanism documented by Massom et al.[3] at additional ice shelf margins.

#### 4.4 Limitations

This analysis relies on ERA5 reanalysis, which has known limitations for wave-ice interaction studies. The wave model does not operate in grid cells with SIC > 30%, preventing direct observation of wave propagation into the ice pack. Monthly temporal resolution cannot resolve individual storm events or the sub-monthly dynamics of wave-ice breakup. The open-water fraction is a crude proxy for fetch, which is properly defined as a directional quantity depending on wind direction and ice geometry. Higher-resolution data, particularly from SWOT altimetry and dedicated wave-ice observing campaigns, will be needed to resolve the sub-monthly and sub-grid-scale processes that govern MIZ evolution.

### 5 Conclusions

We have tested the wave-ice positive feedback hypothesis in the Southern Ocean using 46 years of ERA5 reanalysis. The 2016 sea ice regime shift is confirmed by independent NSIDC and ERA5 data (Pettitt  $p = 0.004$ ). Waves erode the marginal ice zone: SWH Granger-causes MIZ width reduction ( $F = 8.35$ ,  $p < 0.001$ ). However, the feedback loop does not close. Open-water fraction does not amplify wave height ( $p = 0.86$ ), and MIZ erosion does not feed back to total sea ice concentration ( $p = 0.28$ ). Wave intensification across the Southern Ocean is wind-driven ( $p = 0.007$ ), with the SWH change point at 1993, over two decades before the ice shift. The 2016 regime shift exposed more ocean to existing wave energy but did not create additional wave energy through fetch feedback. Among five major ice shelves, only Ross shows significant buffer loss. These results reframe the wave-ice interaction as a wind-mediated, one-directional erosion mechanism rather than a self-reinforcing positive feedback.

### Data Availability

ERA5 reanalysis data are available from the Copernicus Climate Data Store (<https://cds.climate.copernicus.eu>). NSIDC Sea Ice Index v4.0 is available from the National Snow and Ice Data Center (<https://doi.org/10.7265/N5736NV7>). Analysis code is available at <https://github.com/pangeo-data/OpenSCI-Ocean>.

### References

- [1] Claire L. Parkinson. A 40-y record reveals gradual antarctic sea ice increases followed by decreases at rates far exceeding the rates seen in the arctic. *Proceedings of the National Academy of Sciences*, 116(29):14414–14423, 2019.
- [2] Ariaan Purich and Matthew H. England. Compound drivers of Antarctic sea ice loss and Southern Ocean destratification. *Science Advances*, 2026.
- [3] Robert A. Massom, Theodore A. Scambos, Luke G. Bennetts, Phillip Reid, Vernon A. Squire, and Sharon E. Stammerjohn. Antarctic ice shelf disintegration triggered by sea ice loss and ocean swell. *Nature*, 558:383–389, 2018.
- [4] Vernon A. Squire. Ocean wave interactions with sea ice: a reappraisal. *Annual Review of Fluid Mechanics*, 52:37–60, 2020.
- [5] Nander Mokus and Fabien Montiel. Wave-triggered breakup in the marginal ice zone generates lognormal floe size distributions: a simulation study. *The Cryosphere*, 16:4447–4472, 2022.

- [6] Alison L. Kohout, Marcello Vichi, Alberto Alberello, et al. The influence of ocean waves on Antarctic sea-ice albedo and seasonal melting, and potential coupled physical and biological feedbacks. *The Cryosphere*, 20:3271–3290, 2026.
- [7] Alison L. Kohout, Michael J. M. Williams, Shane M. Dean, and Michael H. Meylan. Storm-induced sea-ice breakup and the implications for ice extent. *Nature*, 509:604–607, 2014.
- [8] Ashleigh Womack, Marcello Vichi, Alberto Alberello, and Alessandro Toffoli. Altimetric observation of wave attenuation through the Antarctic marginal ice zone using ICESat-2. *The Cryosphere*, 16:2325–2340, 2022.
- [9] Hans Hersbach, Bill Bell, Paul Berrisford, et al. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146:1999–2049, 2020.
- [10] F. Fetterer, K. Knowles, W. N. Meier, M. Savoie, and A. K. Windnagel. Sea ice index, version 3. *NSIDC: National Snow and Ice Data Center*, 2017.
- [11] A. N. Pettitt. A non-parametric approach to the change-point problem. *Applied Statistics*, 28(2):126–135, 1979.
- [12] C. W. J. Granger. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3):424–438, 1969.
- [13] Gareth J. Marshall. Trends in the Southern Annular Mode from observations and reanalyses. *Journal of Climate*, 16:4134–4143, 2003.